🎓 UniBuddy — The AI University Companion

Your university life, organized.

One platform to manage academic tasks, understand course materials, track academic progress, and get help navigating university life.

AI-powered

Web app

Student-focused

Built in stages

1. The complete feature roadmap

These are the features UniBuddy could eventually have. The release labels below indicate our proposed plan, not features that already exist.

A. Dashboard and Personal Profile

Release 1 — Foundation

The main screen students see after logging in.

- Personalized welcome screen.
- Today's tasks and upcoming deadlines.
- Upcoming exams and assignment reminders.
- Quick access to courses and study materials.
- Overview of completed and pending work.
- User profile with university, department, semester, and preferred language.
- Customizable profile and preferences.

B. University Life Manager

Release 1 — Core functionality

A single place to manage everyday academic responsibilities.

- Add, edit, and delete courses.
- Save class schedules and course information.
- Create assignments, quizzes, and exam entries.
- Set deadlines, priorities, and task status.
- Calendar and timetable views.
- Upcoming-deadline reminders.

- Weekly task overview.
- Personal study goals.
- Mark tasks as completed and review overdue work.

Later improvements: smarter scheduling, automatic study-plan suggestions, and recommendations based on workload.

C. AI Study Buddy

Release 2 — UniBuddy's main AI feature

An AI tutor that helps students understand what they're studying.

- Upload lecture PDFs and course notes.
- Organize materials by subject and course.
- Ask questions about uploaded documents.
- Summarize long lecture notes.
- Explain difficult concepts in simpler language.
- Support Bangla and English explanations.
- Show source-page references for document-based answers.

Later improvements:

- Generate MCQs and practice tests.
- Generate flashcards for revision.
- Create chapter-by-chapter revision guides.
- Explain incorrect quiz answers.
- Recommend which topics to revise next.
- Analyze quiz results to identify weak topics.

D. Academic Performance Tracker

Release 3 — Academic tracking

Tools to help students understand their academic progress.

- GPA and CGPA calculator.
- Semester and course grade records.
- Attendance tracker.
- Track completed courses and academic tasks.
- View academic progress over time.
- Identify subjects requiring more attention.
- Personalized revision recommendations based on available performance data.

The GPA calculator will use explicit grading formulas, while AI can help interpret the results and suggest study priorities.

E. AI University Helpdesk

Release 4 — University information

A place to understand university procedures and official documents.

- Ask questions about university rules and procedures.
- Search supported handbooks, syllabi, notices, and guidelines.
- Summarize lengthy academic notices.
- Explain course-registration and examination instructions.
- Find relevant department contacts and information when available.
- Show document references for answers.
- Clearly indicate when verified information is unavailable.

This module will need a collection of reliable university documents. We should not let the AI invent fees, deadlines, or official rules.

F. Group Projects and Collaboration

Later release — Optional expansion

Features for students working on team assignments and semester projects.

- Create group projects.
- Add team members.
- Assign tasks and responsibilities.
- Track shared deadlines and progress.
- Store project links and notes.
- Use AI to break a large assignment into smaller tasks.
- Generate progress summaries.
- Identify overdue tasks and potential bottlenecks.

G. Supporting Features

These support the entire application and will be introduced alongside the modules that need them.

- Secure registration, login, and logout.
- Private accounts and user-specific data.
- Responsive layouts for desktop, tablet, and mobile browsers.
- Search and filtering across supported records.
- Notification and reminder preferences.

- Loading indicators, validation, and useful error messages.
- Account settings and logout.
- Appropriate control over uploaded files and personal data.
- Reliable storage and recovery of saved work.

2. How we'll build it throughout the semester

This is the proposed development roadmap. We will keep each milestone small enough to complete and demonstrate, rather than introducing every feature immediately.

Stage 0

Planning

Idea, research, and design

Finalize the problem statement, research comparable products, choose the initial technology stack, prepare the proposal, and design the first screens.

Deliverable: proposal, initial UI design, and project repository.

Stage 1

Foundation

Build the basic application

Create the web app structure, registration and login, database, user profile, and first dashboard.

Deliverable: users can log in and access their own dashboard.

Stage 2

Organizer

Make UniBuddy useful without AI

Implement course management, assignments, exam dates, task status, and upcoming deadlines.

Deliverable: a functioning academic organizer with persistent data.

Stage 3

AI release

Introduce the AI Study Buddy

Add PDF uploading, document summaries, and question answering grounded in the student's uploaded materials.

Deliverable: a working AI feature connected to the real application.

Stage 4

Expansion

Add study and tracking tools

Introduce quizzes or flashcards, GPA calculation, and attendance tracking according to team capacity.

Deliverable: additional features demonstrated individually as they are completed.

Stage 5

Polish

Helpdesk, testing, and deployment

Add university-document Q&A if feasible, test existing features, fix bugs, deploy the application, and complete technical documentation.

Deliverable: a stable deployed version, documentation, and presentation-ready demo.

The stages are a product roadmap, not a promise that every later feature will fit into the semester. We can map them to the exact weekly deadlines and deliverables in the course handbook as the plan is finalized. The handbook explicitly emphasizes continuous development, regular GitHub commits, testing, deployment, and documentation.

Software_Development_III_Student_Handbook.pdf

3. What the first version will contain

Let's keep the initial release deliberately focused.

UniBuddy V1 — The first working release

Registration, login, and logout

Student profile

Dashboard with upcoming tasks

Course management

Assignment and exam tracking

Basic deadline reminders

Upload and organize course PDFs

AI document summaries and question answering

Database-backed saved data

Responsive web interface

Use this as a scope checklist. It doesn't mean these features have been implemented yet.

I recommend saving quiz generation, the GPA and attendance tracker, the university helpdesk, and group project management for later releases. They remain part of the UniBuddy long-term vision, without making the first release too large.

4. How we demonstrate continuous progress

Each release should add a real, usable improvement:

- Early weeks: requirements, research, wireframes, database design, and the first UI.
- Middle weeks: functioning accounts, course management, task tracking, and AI integration.
- Later weeks: more study tools, testing, bug fixes, deployment, and documentation.

Push actual work to GitHub regularly, with meaningful commits and a demonstration of each completed milestone. Don't artificially delay finished work just to make it look like the project is progressing. Consistent, verifiable development is the objective.

Software_Development_III_Student_Handbook.pdf

5. The proposed technology structure

Layer	**Proposed choice**
Frontend	React or Next.js
Backend	Node.js with Express
Database	PostgreSQL
AI	An LLM API with document-retrieval support
Authentication	Secure session-based or token-based login
File storage	Private storage for uploaded study materials
Version control	Git and GitHub
Deployment	A hosted frontend, backend, and database

These are initial recommendations, not fixed decisions. The course assignment permits several technology options, including React.js, Next.js, Flutter, Node.js, Django, PostgreSQL, MongoDB, OpenAI, Gemini, and Hugging Face.

Week 01 Assignment - Team Formation and Project Idea - 2026-10-26(1).md

My final recommendation

Keep the full UniBuddy vision, but commit to a small first release. The most important early milestone is a student who can log in, manage courses and deadlines, upload study material, and get useful AI help with that material.

Everything else should be added only after that foundation is reliable. That gives UniBuddy a coherent identity, makes the work manageable for your team, and provides a credible path from the Week 01 idea submission to the final project demonstration.